ARJUNA JEE 2027

Lecture -09 | Mathematics

TRIGONOMETRIC FUNCTIONS

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Topics *to be covered*

A More Formula's

B Cosine Series

C Question Practice

D



LAST CLASS REVISION

What will be the next step if we encounter following during question solving:

(a) $1 + \cos 2\theta = 1 + 2\cos^2\theta - 1 = 2\cos^2\theta$

(b) $1 - \cos 2\theta = 1 - (1 - 2\sin^2\theta) = +2\sin^2\theta$

(c) $\sin\theta\cos\theta = \frac{\sin 2\theta}{2}$

extension

$\frac{1}{2}(2\sin\theta\cdot\cos\theta\cdot\cos 2\theta\cdot\cos 4\theta) = \frac{\sin 8\theta}{8}$

$\frac{1}{2}\times\frac{1}{2}(2\sin 2\theta\cos 2\theta)$

$\frac{1}{4}\times\frac{1}{2}(2\sin 4\theta\cos 4\theta)$

(d) $1 \pm \sin 2\theta = (\sin\theta \pm \cos\theta)^2$



LAST CLASS REVISION

What will be the next step if we encounter following during question solving:

(e) $4\cos^2\theta - 3 = \left(\frac{\cos 3\theta}{\cos\theta}\right)$.

$$\left(4\cos^2\theta - 3\right) \times \frac{\cos\theta}{\cos\theta} = \frac{4\cos^3\theta - 3\cos\theta}{\cos\theta}$$

$$= \frac{\cos 3\theta}{\cos\theta}$$

(f) $4\cos^2\theta - 1 = \frac{\sin 3\theta}{\sin\theta}$.

$$4\left(1 - \sin^2\theta\right) - 1$$

$$4 - 4\sin^2\theta - 1 = \left(3 - 4\sin^2\theta\right) \times \frac{\sin\theta}{\sin\theta}$$



LAST CLASS REVISION

What will be the next step if we encounter following during question solving:

(g) $\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$\frac{1+\cos 2\theta}{2} = 2\cos^2\theta$

- Removal of power 2

(h) $\sin^2\theta = \frac{1-\cos 2\theta}{2}$

$\cos 2\theta = 1 - 2\sin^2\theta$

Question Practice

Question

If $\tan\theta = \frac{3}{4}, \pi < \theta < \frac{3\pi}{2}$, find value of $\sin\frac{\theta}{2}, \cos\frac{\theta}{2}, \cos 2\theta, \tan 2\theta$

III quad.

$$\# \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta} = \frac{2\times\frac{3}{4}}{1-\frac{9}{16}} = \frac{6}{4\left(\frac{7}{16}\right)_4} = \frac{24}{7}$$

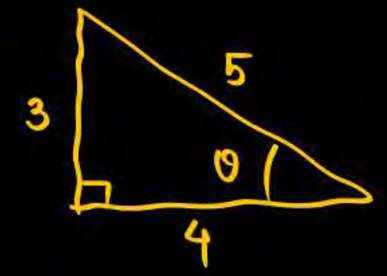
$$\# \sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta} = \frac{24}{25}$$

$$\# \cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta} = \frac{1-\frac{9}{16}}{1+\frac{9}{16}} = \frac{7}{25}$$

Question

If $\tan\theta = \frac{3}{4}, \pi < \theta < \frac{3\pi}{2}$, find value of $\sin\frac{\theta}{2}, \cos\frac{\theta}{2}, \cos 2\theta, \tan 2\theta$

III quad.



$\cos 2\left(\frac{\theta}{2}\right)$

$\sin\theta = \frac{-3}{5}$ & $\cos\theta = \frac{-4}{5}$

$1 - 2\sin^2\frac{\theta}{2} = -\frac{4}{5}$

$\frac{9}{10} = \sin^2\frac{\theta}{2}$

$\frac{9}{5} = 1 + \frac{4}{5} = 2\sin^2\frac{\theta}{2}$

$2\cos^2\frac{\theta}{2} - 1 = -\frac{4}{5}$

$2\cos^2\frac{\theta}{2} = 1 - \frac{4}{5} = \frac{1}{5}$

$\cos^2\frac{\theta}{2} = \frac{1}{10}$

$\cos\frac{\theta}{2} = \pm\sqrt{\frac{1}{10}}$

$\cos\frac{\theta}{2} = -\frac{1}{\sqrt{10}}$

$\pm\frac{3}{\sqrt{10}} = \sin\frac{\theta}{2} = \pm\sqrt{\frac{9}{10}} = \pm\frac{3}{\sqrt{10}}$

$\pi < \theta < \frac{3\pi}{2}$ $\times\frac{1}{2}$

$\frac{\pi}{2} < \frac{\theta}{2} < \frac{3\pi}{4}$

90° 135°



Acidic Question

Show that $\sqrt{2+\sqrt{2+\sqrt{2+2\cos 8\theta}}} = 2\cos\theta$, $8\theta \in$ Iquad. (Given)

$\sqrt{2+2\cos 2\theta}$

$2\cos\theta$.

$2(1+\cos 8\theta)$

$2(1+2\cos^2 4\theta - 1)$

$\sqrt{2+\sqrt{4\cos^2 4\theta}}$

$\sqrt{2+2\cos 4\theta} = \sqrt{4\cos^2 2\theta}$

Question [JEE Mains-2020]

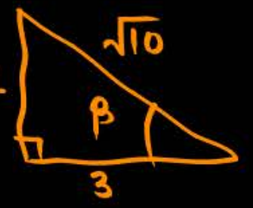
If $\frac{\sqrt{2}\sin\alpha}{\sqrt{1+\cos 2\alpha}} = \frac{1}{7}$ and $\sqrt{\frac{1-\cos 2\beta}{2}} = \frac{1}{\sqrt{10}}$, $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$ (I) then $\tan(\overset{A}{\alpha} + \overset{B}{2\beta})$ is equal to

$$\frac{\sqrt{2}\sin\alpha}{\sqrt{2\cos^2\alpha}} = \frac{1}{7}$$

$$\frac{\sqrt{2}\sin\alpha}{\sqrt{2}\cos\alpha} = \frac{1}{7}$$

$$\tan\alpha = \frac{1}{7}$$

$$\sqrt{\frac{2\sin^2\beta}{2}} = \frac{1}{\sqrt{10}}$$



$$\sin\beta = \frac{1}{\sqrt{10}} \Rightarrow \tan\beta = \pm\frac{1}{3}$$

$$\tan 2\beta = \frac{2\tan\beta}{1-\tan^2\beta} = \frac{2\times\frac{1}{3}}{1-\frac{1}{9}} = \frac{2\times 9}{3\times 8} = \frac{3}{4}$$

$$\frac{\tan\alpha + \tan 2\beta}{1 - \tan\alpha \cdot \tan 2\beta}$$

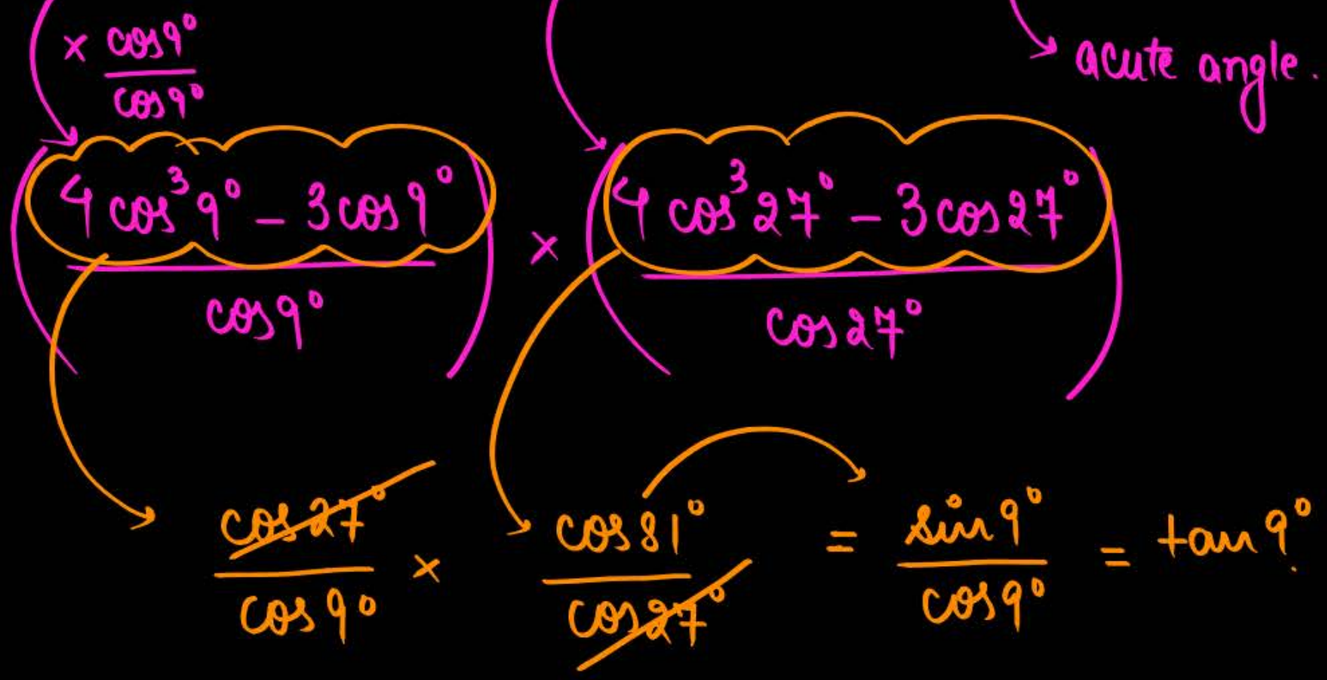
$$\frac{\frac{1}{7} + \frac{3}{4}}{1 - \frac{1}{7}\times\frac{3}{4}} = \frac{4+21}{25}$$

$$= 1$$

Acidic Question

If $(4\cos^2 9^\circ - 3)(4\cos^2 27^\circ - 3) = \tan k^\circ$, then the value of k is _____.

9



76%

Question [JEE Mains-2023]

The value of $36(4\cos^2 9^\circ - 1)(4\cos^2 27^\circ - 1)(4\cos^2 81^\circ - 1)(4\cos^2 243^\circ - 1)$ is



A 18

B 36

C 54

D 27



IMPORTANT VALUES TO MEMORISE

Important Values to Memories :

(1) $\sin 18^\circ = \frac{\sqrt{5}-1}{4} = \cos 72^\circ$

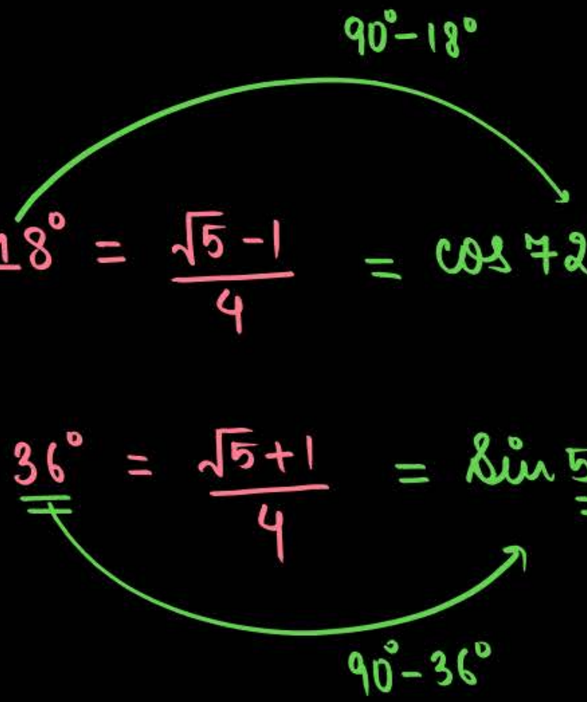
(2) $\sin 54^\circ = \frac{\sqrt{5}+1}{4} = \cos 36^\circ$

(3) $\sin 36^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4} = \cos 54^\circ$

(4) $\sin 72^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4} = \cos 18^\circ$

RAR

$\frac{\pi}{10} = 18^\circ$



- $\sin 18^\circ = \frac{\sqrt{5}-1}{4} = \cos 72^\circ$
- $\cos 36^\circ = \frac{\sqrt{5}+1}{4} = \sin 54^\circ$

for angles less than $45^\circ \rightarrow \sin < \cos$

$\cos\theta = \sqrt{1-\sin^2\theta}$, $\theta \rightarrow$ acute.

$$\cos 18^\circ = \sqrt{1-\sin^2 18^\circ} = \sqrt{1-\left(\frac{\sqrt{5}-1}{4}\right)^2}$$

$$= \sqrt{\frac{16-(5+1-2\sqrt{5})}{16}}$$

$$\cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4} = \sin 72^\circ$$

$\sin 36^\circ = \sqrt{1-\cos^2 36^\circ} = \sqrt{1-\left(\frac{\sqrt{5}+1}{4}\right)^2}$

$$\sin 36^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4} = \cos 54^\circ$$

IMPORTANT VALUES TO MEMORISE

Important Values to Memories :

(5) $\tan\left(\frac{\pi}{8}\right) = \sqrt{2} - 1 = \cot\left(\frac{3\pi}{8}\right)$

RAR

$\frac{\pi}{8} = (22.5)^\circ$

(6) $\tan\left(\frac{3\pi}{8}\right) = \sqrt{2} + 1 = \cot\left(\frac{\pi}{8}\right)$

$\tan\left(\frac{\pi}{2} - \frac{\pi}{8}\right) = \cot\frac{\pi}{8} = \frac{1}{\tan\frac{\pi}{8}} = \frac{1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1} = \sqrt{2}+1$ ✓

Question [JEE Mains-2022]

$\alpha = \sin 36^\circ$ is a root of which of the following equation

A $10x^4 - 10x^2 - 5 = 0$

B $16x^4 - 20x^2 - 5 = 0$

C $16x^4 - 20x^2 + 5 = 0$

D $16x^4 - 10x^2 + 5 = 0$



$$\alpha = \sin 36^\circ = \sqrt{1-\cos^2 36^\circ} = \sqrt{1-\left(\frac{\sqrt{5}+1}{4}\right)^2}$$

$$\alpha = \frac{\sqrt{10-2\sqrt{5}}}{4}$$

$$4\alpha = \sqrt{10-2\sqrt{5}}$$

SBS.

$$16\alpha^2 = 10-2\sqrt{5}$$

$$16\alpha^2 - 10 = -2\sqrt{5}$$

SBS

$$16^2\alpha^4 + 100 - 2(16\alpha^2)(10) = 20$$

$$16^2\alpha^4 - 320\alpha^2 + 80 = 0$$

$\div 16$

$$16\alpha^4 - 20\alpha^2 + 5 = 0$$

Acidic Question

If $\sin 5\theta$ is expanded in increasing power of $\sin\theta$, find all coefficient.

$$\# \quad \sin 5\theta = \sin(\underbrace{3\theta}_{A} + \underbrace{2\theta}_{B}) = \sin 3\theta \cos 2\theta + \cos 3\theta \sin 2\theta$$

$$= (3\sin\theta - 4\sin^3\theta)(1 - 2\sin^2\theta) + (4\cos^3\theta - 3\cos\theta)\, 2\sin\theta\cos\theta$$

$$(4\cos^2\theta - 3)\, 2\sin\theta\cos^2\theta$$

$$(4(1-\sin^2\theta) - 3)\, 2\sin\theta(1-\sin^2\theta)$$

$\sin 5\theta = \sin(3\theta + 2\theta) = \sin 3\theta \cos 2\theta + \cos 3\theta \sin 2\theta$

$= (3\sin\theta - 4\sin^3\theta)(1 - 2\sin^2\theta) + (4\cos^3\theta - 3\cos\theta)\, 2\sin\theta\cos\theta$

$(4\cos^2\theta - 3)\, 2\sin\theta\cos^2\theta$

$(4(1-\sin^2\theta) - 3)\, 2\sin\theta(1-\sin^2\theta)$

$(1 - 4\sin^2\theta)\, 2\sin\theta(1-\sin^2\theta)$

$\sin\theta = t$

$(1-4t^2)\,2t(1-t^2)$

$(3t - 4t^3)(1 - 2t^2) + (2t - 8t^3)(1 - t^2)$

$3t - 4t^3 - 6t^3 + 8t^5 + 2t - 8t^3 - 2t^3 + 8t^5$

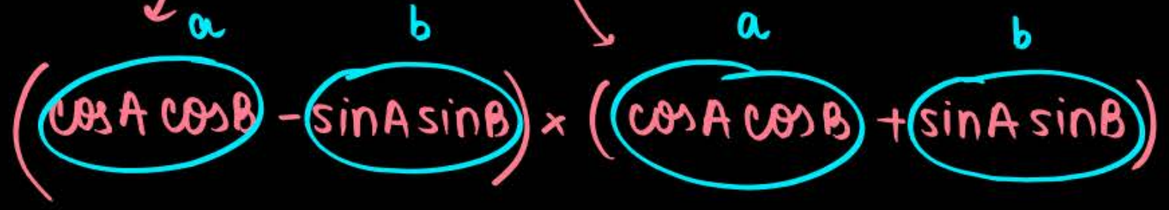
$\sin 5\theta = 16t^5 - 20t^3 + 5t$

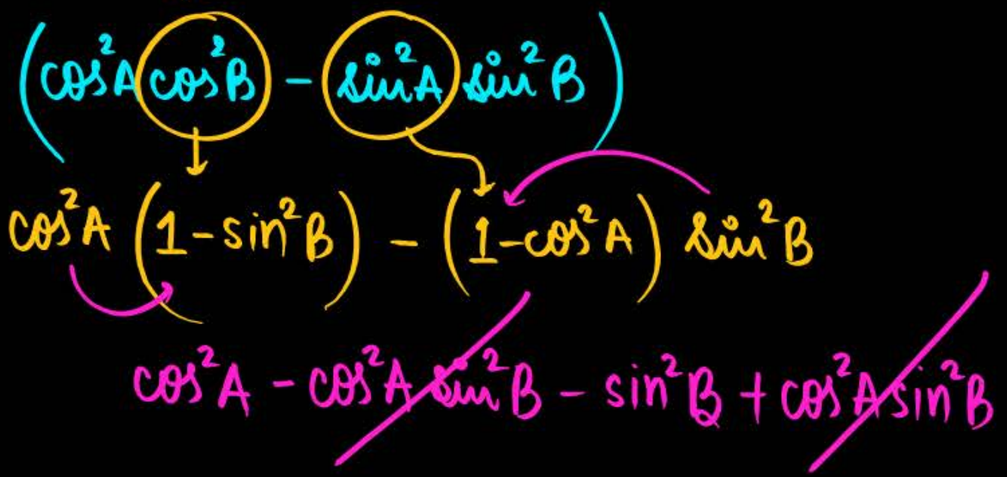
$= 16(\sin\theta)^5 - 20(\sin\theta)^3 + 5(\sin\theta)$

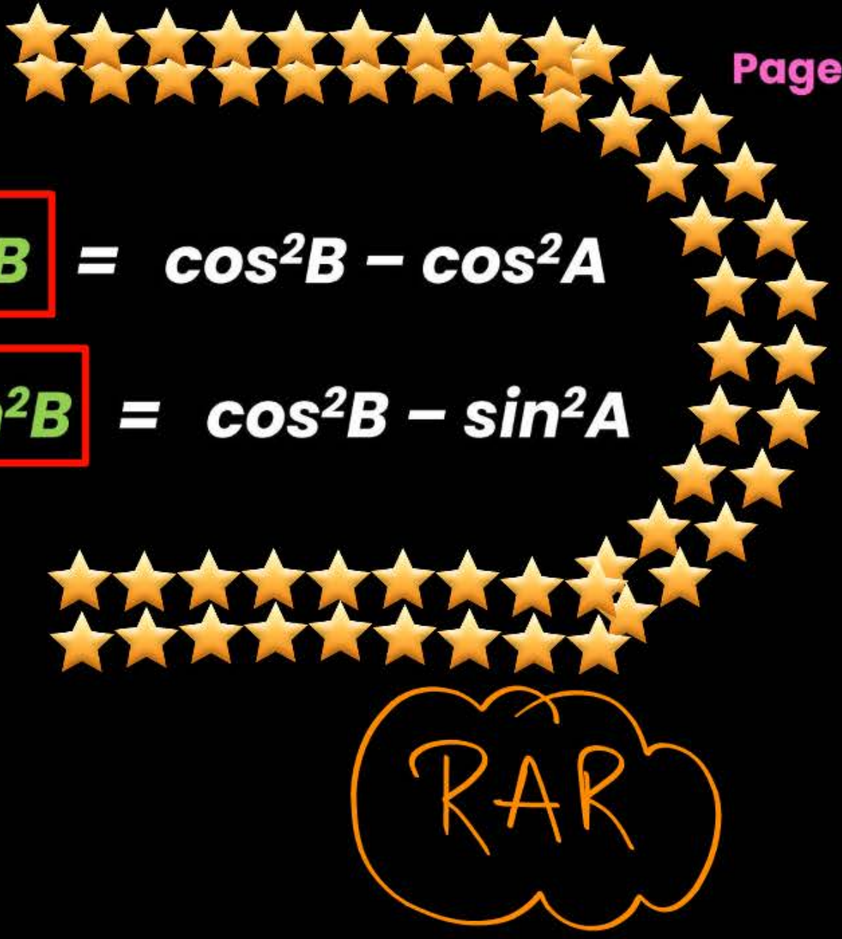
IMPORTANT RESULTS

(i) $\sin(A+B)\cdot\sin(A-B) = \sin^2A - \sin^2B = \cos^2B - \cos^2A$

(ii) $\cos(A+B)\cdot\cos(A-B) = \cos^2A - \sin^2B = \cos^2B - \sin^2A$







RAPID FIRE ROUND

(i) $\sin^2 A - \sin^2 B = \sin(A+B) \cdot \sin(A-B)$

(ii) $\sin^2 A + \sin^2 B =$

(iii) $\cos^2 A + \cos^2 B =$

(iv) $\cos^2 A - \cos^2 B =$

$(1-\sin^2 A) - (1-\sin^2 B)$

$\sin^2 B - \sin^2 A = \sin(B+A)\sin(B-A)$

$1 - \cos^2 A + \sin^2 B$

$1 - (\cos^2 A - \sin^2 B)$

$1 - \cos(A-B)\cdot\cos(A+B)$

$\cos^2 A - \sin^2 B + 1$

IMPORTANT RESULTS

I. $\sin\theta \cdot \sin(60^\circ + \theta) \cdot \sin(60^\circ - \theta) = \frac{1}{4}\sin(3\theta)$

II. $\cos\theta \cdot \cos(60^\circ + \theta) \cdot \cos(60^\circ - \theta) = \frac{1}{4}\cos(3\theta)$

III. $\tan\theta \cdot \tan(60^\circ + \theta) \cdot \tan(60^\circ - \theta) = \tan(3\theta)$

Proof:

$$\sin\theta\left(\sin^2 60^\circ - \sin^2\theta\right)$$

$$\sin\theta\left(\frac{3}{4} - \sin^2\theta\right) = \frac{\sin\theta\left(3 - 4\sin^2\theta\right)}{4} = \frac{3\sin\theta - 4\sin^3\theta}{4} = \frac{\sin 3\theta}{4}$$

IMPORTANT RESULTS

I. $\sin\theta \cdot \sin(60^\circ + \theta) \cdot \sin(60^\circ - \theta) = \frac{1}{4}\sin(3\theta)$

II. $\cos\theta \cdot \cos(60^\circ + \theta) \cdot \cos(60^\circ - \theta) = \frac{1}{4}\cos(3\theta)$

III. $\tan\theta \cdot \tan(60^\circ + \theta) \cdot \tan(60^\circ - \theta) = \tan(3\theta)$

RAR

Question [JEE Mains-2022]

$16\sin(20°)\sin(40°)\sin(80°)$ is equal to

A $\sqrt{3}$

B $2\sqrt{3}$

C 3

D $4\sqrt{3}$

$16\sin(20°)\sin(40°)\sin(80°)$

θ : $20°$, $60°-\theta$: $40°$, $60°+\theta$: $80°$

$= 16 \times \frac{1}{4}\sin(3\times 20°)$

$= 16 \times \frac{1}{4} \times \frac{\sqrt{3}}{2}$

$= 2\sqrt{3}$.

Question [JEE Mains-2019]

The value of $\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ$

θ $\quad \frac{1}{2}$ $\quad 60^\circ - \theta$. $\quad 60^\circ + \theta$.

A $\frac{1}{16}$

B $\frac{1}{36}$

C $\frac{1}{18}$

D $\frac{1}{32}$

$= \frac{1}{2} \times \frac{1}{4} \sin(3 \times 10^\circ)$

$= \frac{1}{8} \times \frac{1}{2} = \frac{1}{16}$.

Acidic Question

Find value of following :

$$\sin 6^\circ \sin 42^\circ \sin 66^\circ \sin 78^\circ =$$

$$\sin 6^\circ \sin(60^\circ+6^\circ)\sin(60^\circ-6^\circ) \quad \sin(60^\circ-18^\circ)\cdot\sin(60^\circ+18^\circ)\sin 18^\circ$$

$$\sin(60^\circ-6^\circ) \qquad \sin 18^\circ$$

$$\frac{1}{4}\frac{\sin(3\times 6^\circ)}{\sin 54^\circ} \times \frac{1}{4}\frac{\sin(3\times 18^\circ)}{\sin 18^\circ} = \frac{1}{16} \text{ Ans.}$$

If $\sin^2(10°)\sin(20°)\sin(40°)\sin(50°)\sin(70°) = \alpha - \frac{1}{16}\sin(10°)$, then $16 + \alpha^{-1}$ is equal to



Maha Bumper Question

If $\alpha = \cot 70^\circ + 4\cos 70^\circ$, $\beta = \sin^2 24^\circ - \cos^2 84^\circ$, $\gamma = \cos^2 73^\circ + \cos^2 47^\circ + \cos 73^\circ \cos 47^\circ$,

$\delta = \sin\frac{\pi}{5}\sin\frac{2\pi}{5}\sin\frac{3\pi}{5}\sin\frac{4\pi}{5}$ → HW

Then find value of : $[\alpha]^2 + [8\beta] + [3\gamma] + [16\delta]$ = (where [.] represents GIF)

$= \frac{\sqrt{5}-1}{8} \Rightarrow 8\beta = \sqrt{5}-1$

$[8\beta] = [\sqrt{5}-1] = 1$

$1 + 1 + 2 +$

$\alpha = \sqrt{3}$

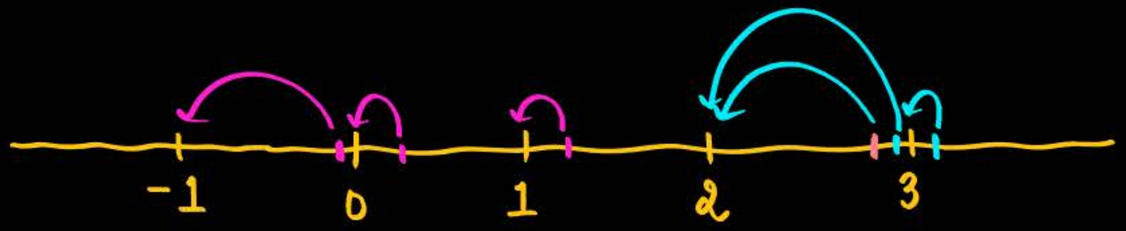
$[\alpha]^2 = [\sqrt{3}]^2 = [1.m]^2 = 1^2 = 1$.

$\gamma = \frac{3}{4} \Rightarrow 3\gamma = \frac{9}{4}$

$[3\gamma] = \left[\frac{9}{4}\right] = 2$.

Note :-

$[x]$ = value is greatest integer less than or equals to x.



$[2.7] = 2$.

$[2.99] = 2$

$[3.01] = 3$.

$[3] = 3$.

$[0.6387] = 0$

$[-0.0001] = -1$

$[-1] = -1$.

$[\sqrt{5} - 1] = 1$

$2.\overline{m} - 1$

$\hookrightarrow 1.\overline{m}$

$$\alpha = \cot 70^\circ + 4\cos 70^\circ$$

$$\alpha = \frac{\cos 70^\circ}{\sin 70^\circ} + 4\cos 70^\circ$$

$$= \frac{\cos 70^\circ + 2(2\cos 70^\circ \sin 70^\circ)}{\sin 70^\circ}$$

$\sin(2\times 70^\circ)$

$$= \frac{\cos 70^\circ + \sin 140^\circ + \sin 140^\circ}{\sin 70^\circ} = \frac{\sin 80^\circ + \sin 140^\circ}{\sin 70^\circ}$$

C D

$$\cos 70^\circ + \sin 140^\circ$$

C D

$$\sin 80^\circ + \sin 140^\circ$$

$$= 2\sin\left(\frac{20^\circ+140^\circ}{2}\right)\cos\left(\frac{20^\circ-140^\circ}{2}\right)$$

$$= 2\sin 80^\circ \cos 60^\circ$$

$$= \frac{2\sin 110^\circ \cos 30^\circ}{\sin 70^\circ} = 2\times\frac{\sqrt{3}}{2} = \sqrt{3}$$

$\sin 70^\circ = \sin(180^\circ - 110^\circ) = \sin 110^\circ$

$$\beta = \sin^2 24^\circ - \cos^2 84^\circ$$

$$\beta = \sin^2 24^\circ - \sin^2 6^\circ$$

$$= \sin(\overbrace{24^\circ + 6^\circ}^{30^\circ}) \sin(\overbrace{24^\circ - 6^\circ}^{18^\circ})$$

$$= \frac{1}{2} \times \frac{\sqrt{5}-1}{4}$$

$$\# \ \beta = \frac{\sqrt{5}-1}{8}$$

$$\beta = \cos^2 66^\circ - \sin^2 6^\circ$$

$$= \cos(\overbrace{66^\circ + 6^\circ}^{72^\circ}) \cos(66^\circ - 6^\circ)$$

$$\beta = \frac{\sqrt{5}-1}{4} \times \frac{1}{2}$$

$\gamma = \cos^2 73^\circ + \cos^2 47^\circ + \cos 73^\circ \cos 47^\circ$

$\cos^2 73^\circ + 1 - \sin^2 47^\circ + \frac{1}{2}\left(2\cos 73^\circ \cos 47^\circ\right)$

$1 + \left(\cos^2 73^\circ - \sin^2 47^\circ\right)$

$\frac{1}{2}\cos(73^\circ + 47^\circ) + \frac{1}{2}\cos(73^\circ - 47^\circ)$

$\Downarrow$

$1 + \cos(73^\circ + 47^\circ)\cos(73^\circ - 47^\circ) + \frac{1}{2}\cos 120^\circ + \frac{1}{2}\cos(73^\circ - 47^\circ)$

$\cos(73^\circ + 47^\circ) = -\frac{1}{2}$

$1 + \frac{1}{2} \times \left(-\frac{1}{2}\right) = 1 - \frac{1}{4} = \frac{3}{4}$

$\# \cos(120^\circ) = \cos(90^\circ + 30^\circ)$

$= -\sin 30^\circ$

$= -\frac{1}{2}$

DO IT BY YOURSELF (DIBY)

DIBY-23

Prove that

1. $\frac{\sin 7\theta-\sin 5\theta}{\cos 7\theta+\cos 5\theta}=\tan\theta$

2. $\frac{\cos 6\theta-\cos 4\theta}{\sin 6\theta+\sin 4\theta}=-\tan\theta$

3. $\frac{\sin A+\sin 3A}{\cos A+\cos 3A}=\tan 2A$

4. $\frac{\sin 7A-\sin A}{\sin 8A-\sin 2A}=\cos 4A\sec 5A$

5. $\frac{\cos 2B+\cos 2A}{\cos 2B-\cos 2A}=\cot(A+B)\cot(A-B)$

6. $\frac{\sin 2A+\sin 2B}{\sin 2A-\sin 2B}=\frac{\tan(A+B)}{\tan(A-B)}$

7. $\frac{\sin A+\sin 2A}{\cos A-\cos 2A}=\cot\frac{A}{2}$

8. $\frac{\sin 5A-\sin 3A}{\cos 3A+\cos 5A}=\tan A$

9. $\frac{\cos 2B-\cos 2A}{\sin 2B+\sin 2A}=\tan(A-B)$

10. $\cos(A+B)+\sin(A-B)=2\sin(45^\circ+A)\cos(45^\circ+B)$

11. $\frac{\cos 3A-\cos A}{\sin 3A-\sin A}+\frac{\cos 2A-\cos 4A}{\sin 4A-\sin 2A}=\frac{\sin A}{\cos 2A\cos 3A}$

DIBY-24

Find the value of expression $\frac{\sin 8\theta\cos\theta - \sin 6\theta\cos 3\theta}{\cos 2\theta\cos\theta - \sin 3\theta\sin 4\theta}$, given $\theta = 7.5°$.

(A) $2+\sqrt{3}$ (B) $2-\sqrt{3}$ (C) 0 (D) 1

[Ans. : (B)]

DIBY-25

If $\alpha = \frac{\pi}{19}$, find the value of $\frac{\sin 23\alpha - \sin 3\alpha}{\sin 16\alpha + \sin 4\alpha}$.

(A) 1 (B) 0 (C) 2 (D) –1

[Ans. : (D)]

DIBY-26

The expression $\frac{(1+\tan 37°)(1+\tan 8°)(1+\tan 15°)(1+\tan 30°)}{(1+\tan 2°)(1+\tan 43°)}$ reduces to

(A) 2 tan 45° (B) 4 (C) 8 (D) 6 cot 45°

[Ans. : (A)]

DIBY-27

$\frac{\sin(\alpha+\beta)}{\sin(\alpha-\beta)} = \frac{2}{3}$ → C&d apply → $\frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{\sin(\alpha+\beta)-\sin(\alpha-\beta)} = \frac{2+3}{2-3}$

For $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, let $3\sin(\alpha+\beta) = 2\sin(\alpha-\beta)$ and a real number k be such that $\tan\alpha = k\tan\beta$. Then the value of k is equal to

(A) 5 (B) −2/3 (C) −5 (D) 2/3

[Ans. : (C)]

DIBY-28

Let $\frac{\tan50^\circ - \tan40^\circ}{\tan10^\circ} = k$, find $k \in N$.

$N^r = \frac{\sin50^\circ}{\cos50^\circ} - \frac{\sin40^\circ}{\cos40^\circ}$

$= \frac{\sin50^\circ\cos40^\circ - \cos50^\circ\sin40^\circ}{\cos50^\circ\cos40^\circ} = \frac{\sin(50^\circ-40^\circ)}{\cos50^\circ\cos40^\circ} = \frac{\sin10^\circ}{\cos50^\circ\cos40^\circ}$

$= \frac{\cos50^\circ\cos40^\circ}{\cos50^\circ\cos40^\circ} + \frac{\sin50^\circ\sin40^\circ}{\cos50^\circ\cos40^\circ} = 1 + \tan50^\circ\tan40^\circ = 2$

[Ans. : 2]

DIBY-29

If $x = \frac{\pi}{48}$, find the value of $\frac{\sin7x+\sin5x+\sin9x+\sin3x}{\cos7x+\cos5x+\cos9x+\cos3x}$.

$\tan50^\circ = \cot40^\circ$

$\tan50^\circ = \frac{1}{\tan40^\circ}$

[Ans. : $\sqrt{2}-1$]

DIBY-30

If $3\sin\alpha = 5\sin\beta \neq 0$, then the value of $\frac{\tan\left(\frac{\alpha+\beta}{2}\right)}{\tan\left(\frac{\alpha-\beta}{2}\right)}$ is:

[Ans. : 4]

DIBY-31

The value of $\cos 24^\circ - \cos 12^\circ + \cos 48^\circ - \cos 84^\circ =$

[Ans. : 1/2]

DIBY-32

If $\cos\alpha\cos\beta = \frac{1}{2}$ and $\sin\alpha\sin\beta = \frac{1}{3}$

Column - I	Column - II
(A) $\cos\frac{\alpha+\beta}{2}$	(P) $\pm\frac{\sqrt{13}}{12}$
(B) $\cos\frac{\alpha-\beta}{2}$	(Q) $\frac{2}{3}$
(C) $\tan\left(\frac{\alpha+\beta}{2}\right)$	(R) $\pm\frac{3}{\sqrt{13}}$
(D) $\tan\left(\frac{\alpha-\beta}{2}\right)$	(S) $\pm\sqrt{\frac{131}{13}}$

[Ans. : A-R,B-P,C-Q,D-S]

DIBY-33

Prove that $\frac{\sin\theta+\sin 2\theta}{1+\cos\theta+\cos 2\theta}=\tan\theta$

DIBY-34

Prove that $\frac{1-\cos A+\cos B-\cos(A+B)}{1+\cos A-\cos B-\cos(A+B)}=\tan\frac{A}{2}\cot\frac{B}{2}$

DIBY-35

Value of : $\prod_{k=1}^{17}\sin\frac{k\pi}{18}=\frac{a}{2^{16}}$ then $a=$ _____.

$\sin\frac{\pi}{18}\cdot\sin\frac{2\pi}{18}\cdot\sin\frac{3\pi}{18}\cdots\sin\frac{17\pi}{18}$

[Ans. : 9]

DIBY-36

If $\frac{5\pi}{2}<x<3\pi$, then the value of the expression $\frac{\sqrt{1-\sin x}+\sqrt{1+\sin x}}{\sqrt{1-\sin x}-\sqrt{1+\sin x}}$ is

(A) $-\cot\frac{x}{2}$ (B) $\cot\frac{x}{2}$ (C) $\tan\frac{x}{2}$ (D) $-\tan\frac{x}{2}$

[Ans. : (C)]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians